A Note on Twisted Jordan Derivations of Incidence Algebras

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ARTICLE HISTORY
Preprint version 1, compiled 2026/10/17

ABSTRACT
Let A be the incidence algebra of a finite partially ordered set X (a poset) over a commutative ring K in which 2 is invertible. Let τ be an order automorphism of X with $\tau^2 = \mathrm{id}$, and let α be the automorphism of A that τ induces. We twist the Jordan product of A by α, and we call a linear map that satisfies the Leibniz rule for this product a braided-symmetric derivation. We prove that the braided-symmetric derivations are exactly the derivations that commute with α. The commuting condition is not an assumption, because every braided-symmetric derivation satisfies it. For $\tau = \mathrm{id}$ this is the known theorem that Jordan derivations of incidence algebras are derivations. We show that the hypothesis on 2 is needed, and we list computed dimensions for small posets.

1. Introduction

A linear map D on an associative algebra A is a derivation if $D(xy) = D(x)y + xD(y)$ for all $x, y \in A$. It is a Jordan derivation if the same rule holds for the product $xy + yx$. The standard question is whether every Jordan derivation is a derivation [4]. For incidence algebras of locally finite preordered sets, Xiao proved that the answer is yes when the base ring is 2-torsion free [4, Theorem 3.3]. Zhang and Khrypchenko proved that every Lie derivation of the incidence algebra of a connected locally finite preordered set, over a 2-torsion free ring, is the sum of a derivation and a map into the centre [7, Theorem 3.10]. Yang described the nonlinear Jordan derivations of incidence algebras over a 2-torsion free commutative ring [5, Theorem 3.1]. Yang also studied Jordan $*$-derivations, where $*$ is an involution that reverses products, and showed that such maps need not be $*$-derivations [6, Example 3.1]. Jia and Xiao proved that every Jordan skew derivation of an incidence algebra over a field of characteristic not 2 is a skew derivation [3, Theorem 3.8]. In the skew results the automorphism acts in the Leibniz rule [2,3]. Here it acts in the product, which is a different problem. Fornaroli and Khrypchenko described the skew derivations of incidence algebras, that is, the φ-derivations for an automorphism φ [2, Theorem 2.8].

In an earlier paper we studied derivation-like maps for a product that is twisted by an involution of $A \otimes A$ [1, Section 3]. We call them braided-symmetric derivations. The twist by an involutive automorphism α of A replaces $xy + yx$ by $xy + \alpha(yx)$ [1]. Here we answer the question for this product on incidence algebras.

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In this paper we use the following set-up. Let K be a commutative ring in which 2 is invertible. Let X be a finite poset and let $A = I(X, K)$ be its incidence algebra over K, defined in Section 2. Let τ be an order automorphism of X with $\tau^2 = \mathrm{id}$. It induces an involutive automorphism α of A, as shown in Section 2. Write $x \cdot y = xy + \alpha(yx)$ for $x, y \in A$. A linear map $D\colon A \to A$ is braided-symmetric if $D(x \cdot y) = D(x) \cdot y + x \cdot D(y)$ for all x, y. Let $\mathrm{BSDer}(A, \alpha)$ be the set of these maps, and let $\mathrm{BSDer}(A, \alpha)^\alpha$ be the subset of maps that commute with α. Let $\mathrm{Der}(A)^\alpha$ be the set of derivations of A that commute with α.

Theorem 1.1. *Let K be a commutative ring with 2 invertible, let X be a finite poset, and let τ be an order automorphism of X with $\tau^2 = \mathrm{id}$. Let $A = I(X, K)$ and let α be the automorphism of A induced by τ. Then $\mathrm{BSDer}(A, \alpha) = \mathrm{Der}(A)^\alpha$. In particular every braided-symmetric derivation commutes with α.*

For $\tau = \mathrm{id}$ the map α is the identity and $x \cdot y = xy + yx$. Then Theorem 1.1 says that a Jordan derivation of $I(X, K)$ is a derivation. This case is Theorem 1.2, a theorem of Xiao [4, Theorem 3.3], which we recall.

In his paper, X is a locally finite preordered set [4, p. 1357]. Our poset X is finite, so the element c of Step 1 is a finite sum.

Theorem 1.2 ([4, Theorem 3.3])**.** *Let R be a 2-torsion free commutative ring with identity. Then every Jordan derivation of the incidence algebra $I(X, R)$ is a derivation.*

So the new content is the twist. The proof uses the diagonal idempotents of A and no description of its derivations.

Section 2 fixes notation and gives a criterion for membership in $\mathrm{BSDer}(A, \alpha)$. Section 3 proves Theorem 1.1. Section 4 shows that 2 must be invertible and lists dimensions computed by computer.

2. Preliminaries

Throughout, K is a commutative ring with 2 invertible, X is a finite poset, and τ is an order automorphism of X with $\tau^2 = \mathrm{id}$. Write $x \le y$ for the order of X.

The incidence algebra of a finite poset is the algebra defined by Xiao [4, p. 1357].

Definition 2.1. The incidence algebra $A = I(X, K)$ is the free K-module with basis $\{e_{xy} : x \le y\}$. The product is $e_{xy}e_{zw} = e_{xw}$ if $y = z$ and 0 otherwise. Write $\varepsilon_x = e_{xx}$. The elements ε_x are orthogonal idempotents and $1 = \sum_x \varepsilon_x$.

For $x \le y$ we have $\tau x \le \tau y$, so the K-linear map α with $\alpha(e_{xy}) = e_{\tau x\, \tau y}$ is defined. It is an automorphism of A and $\alpha^2 = \mathrm{id}$. Note that $\alpha(\varepsilon_x) = \varepsilon_{\tau x}$.

Lemma 2.2. *If $x \le \tau x$ then $x = \tau x$.*

Proof. Apply τ to $x \le \tau x$. We get $\tau x \le \tau^2 x = x$. So $x = \tau x$ because $\le$ is antisymmetric. □

Let $D\colon A \to A$ be K-linear. Its Leibniz defect is the bilinear map

$$E_D(x, y) = D(xy) - D(x)y - xD(y).$$

So D is a derivation if and only if $E_D = 0$.

Lemma 2.3. *Let $D\colon A \to A$ be linear with $D\alpha = \alpha D$. Then $D \in \mathrm{BSDer}(A,\alpha)$ if and only if $E_D(x,y) + \alpha(E_D(y,x)) = 0$ for all $x, y \in A$. In particular every derivation that commutes with α lies in $\mathrm{BSDer}(A,\alpha)^\alpha$.*

Proof. Write $E = E_D$. Since D commutes with α, we have $\alpha(y)\,\alpha(D(x)) = \alpha(yD(x))$. So

$$D(x)\cdot y + x\cdot D(y) = D(x)y + xD(y) + \alpha\big(yD(x) + D(y)x\big).$$

By the definition of E, the right side equals

$$D(xy) - E(x,y) + \alpha\big(D(yx) - E(y,x)\big).$$

The left side of the Leibniz rule is $D(x\cdot y) = D(xy) + \alpha(D(yx))$. So the Leibniz rule holds exactly when $E(x,y) + \alpha(E(y,x)) = 0$. If D is a derivation then $E = 0$ and this holds. □

The next lemma shows that the commuting condition holds for every braided-symmetric derivation.

Lemma 2.4. *Let $D \in \mathrm{BSDer}(A,\alpha)$. Then $D(1) = 0$ and $D\alpha = \alpha D$. So $\mathrm{BSDer}(A,\alpha) = \mathrm{BSDer}(A,\alpha)^\alpha$.*

Proof. We have $1\cdot 1 = 1 + \alpha(1) = 2$, so $D(1\cdot 1) = 2D(1)$. Also $D(1)\cdot 1 = D(1) + \alpha(D(1)) = 1\cdot D(1)$. The Leibniz rule gives $D(1\cdot 1) = 2D(1) + 2\alpha(D(1))$. So $2\alpha(D(1)) = 0$, and since 2 is invertible and α is injective, $D(1) = 0$. Now let $x \in A$. Then $1\cdot x = x + \alpha(x)$ and $D(1)\cdot x = 0$, so the Leibniz rule gives $D(x) + D(\alpha(x)) = D(1)\cdot x + 1\cdot D(x) = D(x) + \alpha(D(x))$. Hence $D(\alpha(x)) = \alpha(D(x))$. □

The proof uses only the unit of A, the invertibility of 2 and the injectivity of α. So Lemma 2.4 holds for every unital K-algebra with an automorphism α, when 2 is invertible in K.

By Lemma 2.3, the inclusion $\mathrm{Der}(A)^\alpha \subseteq \mathrm{BSDer}(A,\alpha)$ is clear. By Lemma 2.4, it remains to show that a map D in $\mathrm{BSDer}(A,\alpha)^\alpha$ is a derivation.

3. Proof of the theorem

Fix $D \in \mathrm{BSDer}(A,\alpha)^\alpha$ and write $E = E_D$. By Lemma 2.3,

$$E(x,y) = -\alpha\big(E(y,x)\big) \qquad (x,y \in A). \tag{1}$$

For $x \in X$ and $r \le s$, let $F_x(r,s) \in K$ be the coefficient of e_{rs} in $D(\varepsilon_x)$. Since $D\alpha = \alpha D$ and $\alpha(\varepsilon_x) = \varepsilon_{\tau x}$,

$$F_{\tau x}(\tau r, \tau s) = F_x(r,s). \tag{2}$$

Let $x \neq y$ in X and put $P_{xy} = D(\varepsilon_x)\varepsilon_y + \varepsilon_x D(\varepsilon_y)$. Since $\varepsilon_x\varepsilon_y = 0$, we have $E(\varepsilon_x,\varepsilon_y) = -P_{xy}$. By (1) and $\alpha(P_{yx}) = P_{\tau y\,\tau x}$ we get

$$P_{xy} + P_{\tau y\,\tau x} = 0 \qquad (x \neq y). \tag{3}$$

The coefficient of e_{rs} in P_{xy} is $\delta_{sy}F_x(r,y) + \delta_{rx}F_y(x,s)$. So the coefficient of e_{rs} in (3) is

$$\delta_{sy}F_x(r,y) + \delta_{rx}F_y(x,s) + \delta_{s,\tau x}F_{\tau y}(r,\tau x) + \delta_{r,\tau y}F_{\tau x}(\tau y,s) = 0. \tag{4}$$

Lemma 3.1. *Let $r \le s$ and suppose $F_x(r,s) \neq 0$. Then $r = x$ or $s = x$ or $r = s$. In the last case $r = \tau r$ or $r = \tau x$.*

Proof. Suppose $r \neq x$ and $s \neq x$. Put $y = s$ in (4). This is allowed since $s \neq x$. The second term needs $r = x$, so it is zero. So if also $s \neq \tau x$ and $r \neq \tau s$, the first term alone gives $F_x(r,s) = 0$. Hence $s = \tau x$ or $r = \tau s$.

Now put $(x', y') = (r, x)$ in (4), which is allowed since $r \neq x$, and take the coefficient at (r,s). The first term needs $s = x$, so it is zero. The second term is $F_x(r,s)$. If $s \neq \tau r$ and $r \neq \tau x$, the other two terms are zero and $F_x(r,s) = 0$. Hence $s = \tau r$ or $r = \tau x$.

Combine the two conclusions. If $r = \tau s$, then $s = \tau r$, and $r \le s = \tau r$ gives $r = \tau r = s$ by Lemma 2.2. Otherwise $s = \tau x$, and then $s = \tau r$ would give $r = x$, which is excluded. So $r = \tau x$, and $r = s = \tau x$. □

So $D(\varepsilon_x)$ is a sum of terms e_{xs} with $s > x$, terms e_{rx} with $r < x$, and diagonal terms.

Step 1: removing an inner derivation For $r < s$ put $c_{rs} = F_s(r,s)$ and let $c = \sum_{r<s} c_{rs}e_{rs}$. By (2) and since τ maps the set of pairs $r < s$ onto itself, we have $\alpha(c) = c$. The commutator with ε_x is

$$[c, \varepsilon_x] = \sum_{r<x} c_{rx}e_{rx} - \sum_{s>x} c_{xs}e_{xs}.$$

The coefficient of e_{rx} for $r < x$ equals $F_x(r,x)$ by the definition of c. For $x < s$, we claim that $F_x(x,s) = -F_s(x,s)$. Indeed, put $(r,s) = (x,s)$ and $y = s$ in (4). The first two terms are $F_x(x,s) + F_s(x,s)$. The third term needs $s = \tau x$ and the fourth needs $x = \tau s$. Both would give $x < \tau x$, which contradicts Lemma 2.2. So the claim holds. Hence $[c, \varepsilon_x]$ is exactly the off-diagonal part of $D(\varepsilon_x)$.

Let $D_1 = D - \mathrm{ad}_c$. Since $\alpha(c) = c$, the inner derivation ad_c commutes with α. So D_1 commutes with α and $E_{D_1} = E_D$. By Lemma 2.3, $D_1 \in \mathrm{BSDer}(A,\alpha)^\alpha$, and D is a derivation if and only if D_1 is. By construction each $D_1(\varepsilon_x)$ is diagonal:

$$D_1(\varepsilon_x) = \sum_y d_x(y)\,\varepsilon_y, \qquad d_{\tau x}(\tau y) = d_x(y).$$

Relation (1) still holds for D_1.

Step 2: D_1 vanishes on the idempotents Let $x \neq y$ and put $p = d_x(y)$ and $q = d_y(x)$. Then $P_{xy} = p\varepsilon_y + q\varepsilon_x$ and $P_{\tau y\,\tau x} = q\varepsilon_{\tau x} + p\varepsilon_{\tau y}$, so (3) reads

$$p\,(\varepsilon_y + \varepsilon_{\tau y}) + q\,(\varepsilon_x + \varepsilon_{\tau x}) = 0.$$

Suppose first that $\{x, \tau x\} \neq \{y, \tau y\}$. Then the two sums have no idempotent in common. Each sum is either twice one idempotent or the sum of two different idempotents. Since 2 is invertible, $p = q = 0$. Suppose now that $\{x, \tau x\} = \{y, \tau y\}$. Since $x \neq y$,

we have $y = \tau x \neq x$. The relation becomes $(p+q)(\varepsilon_x + \varepsilon_{\tau x}) = 0$, so $p + q = 0$. But $q = d_{\tau x}(x) = d_x(\tau x) = p$ by the symmetry of d. So $2p = 0$ and $p = 0$. Hence $d_x(y) = 0$ for $x \neq y$, and $D_1(\varepsilon_x) = \lambda_x \varepsilon_x$ for a scalar λ_x.

Now $E(\varepsilon_x, \varepsilon_x) = \lambda_x \varepsilon_x - 2\lambda_x \varepsilon_x = -\lambda_x \varepsilon_x$. Relation (1) gives $-\lambda_x \varepsilon_x = \lambda_x \varepsilon_{\tau x}$. If $\tau x \neq x$ this gives $\lambda_x = 0$. If $\tau x = x$ it gives $2\lambda_x = 0$, so $\lambda_x = 0$. So $D_1(\varepsilon_x) = 0$ for all x.

Step 3: D_1 on the other basis elements Let $c_0 < d_0$ in X and write $H = D_1(e_{c_0 d_0})$, so that $\alpha(H) = D_1(e_{\tau c_0\, \tau d_0})$. Take $x \in X$. Since $D_1(\varepsilon_x) = 0$, we have $E(\varepsilon_x, e_{c_0 d_0}) = \delta_{x c_0} H - \varepsilon_x H$ and $E(e_{c_0 d_0}, \varepsilon_x) = \delta_{x d_0} H - H \varepsilon_x$. Relation (1) for the pair $(\varepsilon_x, e_{c_0 d_0})$ gives

$$\delta_{x c_0} H - \varepsilon_x H + \delta_{x d_0} \alpha(H) - \alpha(H) \varepsilon_{\tau x} = 0. \tag{5}$$

Let $x \notin \{c_0, d_0\}$. Then $\varepsilon_x H = -\alpha(H)\varepsilon_{\tau x}$. The left side lies in $\varepsilon_x A$ and the right side lies in $A\varepsilon_{\tau x}$. So both lie in $\varepsilon_x A \varepsilon_{\tau x}$, which is spanned by $e_{x\,\tau x}$ if $x \leq \tau x$ and is zero otherwise. By Lemma 2.2, it is zero unless $\tau x = x$. If $\tau x = x$, the common value is $h\,\varepsilon_x$. Applying α to $\varepsilon_x H = h \varepsilon_x$ gives $\varepsilon_x \alpha(H) = h \varepsilon_x$, while $\alpha(H)\varepsilon_x = -h\varepsilon_x$. So the coefficient of ε_x in $\alpha(H)$ is both h and $-h$. So $h = 0$. Hence $\varepsilon_x H = 0$ and $\alpha(H)\varepsilon_{\tau x} = 0$, that is, $H\varepsilon_x = 0$, for every $x \notin \{c_0, d_0\}$. So H is supported on the entries with row and column in $\{c_0, d_0\}$. Since $c_0 < d_0$,

$$H = p\,\varepsilon_{c_0} + q\,e_{c_0 d_0} + r\,\varepsilon_{d_0}.$$

Put $x = c_0$ in (5). Then $(1 - \varepsilon_{c_0})H = \alpha(H)\varepsilon_{\tau c_0}$, that is, $r\,\varepsilon_{d_0} = p\,\varepsilon_{\tau c_0}$. We have $d_0 \neq \tau c_0$, since $c_0 < d_0 = \tau c_0$ would contradict Lemma 2.2. So $r = p = 0$ and $D_1(e_{c_0 d_0}) = q_{c_0 d_0}\, e_{c_0 d_0}$ for a scalar $q_{c_0 d_0}$.

Step 4: D_1 is a derivation Put $q_{xx} = 0$. Then $D_1(e_{xy}) = q_{xy} e_{xy}$ for all $x \leq y$. Let $a < b < d$. Then $e_{bd} e_{ab} = 0$, so $E(e_{bd}, e_{ab}) = 0$. By (1), $E(e_{ab}, e_{bd}) = 0$, that is, $q_{ad} = q_{ab} + q_{bd}$. This holds if two of the indices coincide, since then one of the terms is zero. Now check $E = 0$ on all pairs of basis elements. For e_{xy} and e_{zw} with $y \neq z$ we get $D_1(0) - q_{xy} e_{xy} e_{zw} - q_{zw} e_{xy} e_{zw} = 0$. For $y = z$ we get $(q_{xw} - q_{xy} - q_{yw}) e_{xw} = 0$. So $E_{D_1} = 0$ and D_1 is a derivation. Hence $D = D_1 + \mathrm{ad}_c$ is a derivation. By Lemma 2.4, D commutes with α, so $D \in \mathrm{Der}(A)^\alpha$. This proves Theorem 1.1. □

Remark 3.2. The proof uses that 2 is invertible in four places: to cancel $2\alpha(D(1)) = 0$ in Lemma 2.4, $2p = 0$ and $2\lambda_x = 0$ in Step 2, and $2h = 0$ in Step 3.

4. Examples and sharpness

The hypothesis that 2 is invertible is needed.

Example 4.1. Let $X = \{1, 2\}$ be a set of two incomparable points with τ the swap, and let $K = \mathbb{F}_2$. Then $A = \mathbb{F}_2 \times \mathbb{F}_2$. The linear map D with $D(\varepsilon_1) = D(\varepsilon_2) = \varepsilon_1 + \varepsilon_2$ commutes with α. Here $x \cdot y = xy + \alpha(yx)$ equals $xy + \alpha(xy)$, and both sides of the Leibniz rule vanish on the basis. So $D \in \mathrm{BSDer}(A, \alpha)^\alpha$. But a derivation satisfies $D(\varepsilon_1) = D(\varepsilon_1^2) = 2\varepsilon_1 D(\varepsilon_1) = 0$ in characteristic 2, while $D(\varepsilon_1) \neq 0$. So D is not a derivation.

The next table lists dimensions computed by computer from the rank of the linear system that defines $\mathrm{BSDer}(A,\alpha)^\alpha$ and $\mathrm{Der}(A)^\alpha$. Each entry is the pair $(\dim \mathrm{BSDer}(A,\alpha)^\alpha,\ \dim \mathrm{Der}(A)^\alpha)$ over the field $\mathbb{F}_p$. Since $\mathrm{BSDer}(A,\alpha)^\alpha = \mathrm{BSDer}(A,\alpha)$ by Lemma 2.4 and $\mathrm{Der}(A)^\alpha \subseteq \mathrm{BSDer}(A,\alpha)$ by Lemma 2.3, equal dimensions mean equal spaces.

poset X	τ	$p=2$	$p=3$
2 incomparable points	swap	$(1,0)$	$(0,0)$
chain of 2 points	identity	$(4,2)$	$(2,2)$
chain of 3 points	identity	$(8,5)$	$(5,5)$
V: $0<1$, $0<2$	identity	$(7,4)$	$(4,4)$
V	swap $1 \leftrightarrow 2$	$(3,2)$	$(2,2)$
diamond $0<1,2<3$	swap $1 \leftrightarrow 2$	$(6,5)$	$(5,5)$
cross: $0,1<2<3,4$	swap $0 \leftrightarrow 1$, $3 \leftrightarrow 4$	$(6,6)$	$(6,6)$

For $p=3$ the dimensions agree in every row, as Theorem 1.1 predicts. For $p=2$ they differ in all rows but the last. So in characteristic 2 the conclusion can hold or fail, depending on (X,τ). We do not know which pairs (X,τ) keep equality in characteristic 2.

Remark 4.2. The space $\mathrm{Der}(A)^\alpha$ is in general different from the space of α-derivations of Fornaroli and Khrypchenko [2]. An α-derivation d satisfies $d(xy) = d(x)\alpha(y) + \alpha(x)d(y)$, which is not the Leibniz rule for $x \cdot y$.

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